

Rina Matsumoto

Senior ML Engineer – Infrastructure & Developer Tools

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CORE COMPETENCIES

ML/AI: Retrieval-Augmented Generation (RAG), multi-agent frameworks, LLM fine-tuning, prompt engineering, vector databases, semantic search, agentic reasoning loops, function calling, tool use orchestration

Frameworks & Libraries: LangChain, LlamaIndex, AutoGen, Pydantic, OpenAI API, Anthropic API, Cohere, HuggingFace Transformers, vLLM, Ray, FastAPI

Infrastructure & Tools: Docker, Kubernetes, PostgreSQL, Pinecone, Weaviate, Redis, Apache Kafka, MLflow, Weights & Biases, GitHub Actions

Evaluation & Monitoring: RAGAS framework, custom eval harnesses, benchmark construction, latency profiling, token cost optimization

Languages: Python, TypeScript, SQL, Bash

PROFESSIONAL EXPERIENCE

Senior ML Engineer – Developer Experience

Tessera Labs, Portland, OR | Mar 2024 – Present

- Architected and maintained an open-source LLM toolkit for building multi-agent applications, now at 4.2K GitHub stars and 180+ contributors; designed TypeScript and Python SDKs with unified abstractions for agent state management and tool execution
- Built comprehensive evaluation framework for agent reasoning quality, implementing RAGAS metrics and custom recall/precision benchmarks; measured 87% semantic correctness on internal eval set of 500 queries against baseline of 61%
- Developed RAG infrastructure component supporting multiple retrieval strategies (dense, sparse, hybrid); optimized vector search latency to <45ms p99 for 10M-token corpus using Weaviate clustering and query rewriting
- Designed and shipped fine-tuning pipeline orchestration using Ray distributed training; reduced wall-clock time for instruction-tuned models from 8 hours to 2.5 hours on 4xA100 setup through batch scheduling improvements
- Led technical blog series on agentic reasoning patterns, published five posts on agent memory architectures, tool-use optimization, and LLM routing strategies; accumulated 12K+ views and cited by community projects

ML Infrastructure Engineer

Zenith AI, Seattle, WA | Jul 2023 – Feb 2024

- Owned development of internal LLM serving infrastructure; built multi-model endpoint abstraction supporting OpenAI, Anthropic, and open-weight model inference with automatic rate-limit handling and request queuing via Kafka
- Implemented function-calling orchestration layer enabling tool composition across 40+ API integrations; benchmarked at 78% tool selection accuracy on held-out test set of 200 complex multi-step tasks
- Constructed evaluation harness for LLM output quality using custom metrics (BLEU, semantic similarity via embeddings, token-level accuracy); integrated with CI/CD to track model performance across versions
- Optimized prompt templates and context management strategies, reducing average token usage by 32% across production query workloads while maintaining ROUGE-L score of 0.71 on benchmark tasks

- Mentored three junior engineers on LLM fundamentals, vector database design, and debugging agentic failures; conducted internal workshop on RAG system architecture

ML Engineer

Cascade AI, Eugene, OR | Jan 2023 – Jun 2023

- Developed end-to-end RAG system for technical documentation retrieval using LlamaIndex, PostgreSQL with pgvector, and semantic chunking; achieved 83% MRR@10 on 5K-document test collection
- Fine-tuned open-weight LLMs (Llama 2, Mistral) on domain-specific instruction sets using LoRA; validated improvements through offline benchmarks showing 14-point increase in task-specific accuracy
- Built agent framework supporting multi-turn conversation state and tool memory; implemented reasoning chain evaluation using custom scoring rubric applied to 300-example validation set
- Contributed to internal MLflow tracking infrastructure, implementing automated experiment logging and model registry workflows; reduced setup friction for team onboarding

EDUCATION

M.S. Computer Science – University of Washington, Seattle, WA | 2022

Focus: Machine Learning, Natural Language Processing

B.S. Computer Science – Reed College, Portland, OR | 2020

TECHNICAL PUBLICATIONS & COMMUNITY

- "Building Reliable Tool-Use Agents" – Published on Towards Data Science (Jul 2025)
- Speaker, LLM Infrastructure Summit – "Observability Patterns for Multi-Agent Systems" (Nov 2025)
- Core contributor to LlamaIndex framework; PRs focused on hybrid retrieval and prompt optimization